

Pregnant Cardiomyopathy



Section I: Scenario Demographics

Scenario Title:	Pregnant Cardiomyopathy
Date of Development:	(16/04/2018)
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Nadia Primiani and Sev Perelman
Affiliations/Institution(s):	Mount Sinai Hospital
Contact E-mail (optional):	

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To manage acute CHF and cardiogenic shock in a pregnant patient with cardiomyopathy
CRM Objectives:	1. Effectively lead a team through the uncommon presentation of cardiogenic shock in pregnancy 2. Efficiently utilize closed-loop communication between team members 3. Communicate with consultants early in the management of a sick pregnant patient
Medical Objectives:	1) Recognize a critically ill pregnant patient with cardiogenic shock 2) Work through the differential diagnosis of dyspnea in pregnancy 3) Facilitate safe intubation with considerations for a pregnant patient

Case Summary: Brief Summary of Case Progression and Major Events

A 38-year-old female G2P1 at 36 weeks GA presents with acute on chronic respiratory distress in addition to chronic peripheral edema. She undergoes respiratory fatigue and hypoxia requiring intubation. She then becomes hypotensive which the team discovers is secondary to cardiogenic shock, requiring vasopressor infusion and consultation with Cardiology/ICU.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- Chris Nickson. (2013). Peripartum Cardiomyopathy. Accessed April 27, 2018 from <https://lifeinthefastlane.com/ccp/peripartum-cardiomyopathy/>



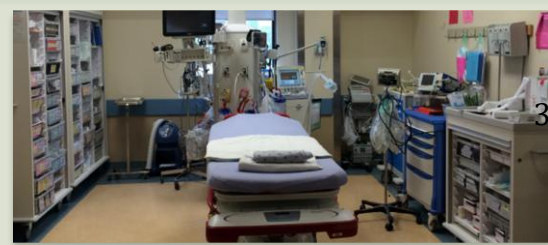
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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working in a community ED and your team has been called urgently by the nurse to see a 38 year old female who is G2P1 at 36 weeks gestational age. She was brought in by her sister, who is quite agitated and upset, saying “everybody has been ignoring her symptoms for the last 4 weeks”. The patient has just experienced a syncopal episode at home.			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates	Brief Description of Role		
Nurse	To assist at bedside, cue team to patient’s status PRN		
RT	To assist with intubation		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☒ Arterial Line	☒ Other: Fetal Doppler	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☒ LMA	☐ Other:	
E. Moulage			
Pregnant belly			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	20 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History							
Patient Name: Sunny Mills			Age: 38		Weight: 75 kg		
Gender: ☐ M ☒ F			Code Status: Full				
Chief Complaint: Syncopal episode, now difficulty breathing							
History of Presenting Illness: Aching legs with edema x 4 weeks. Mildly dyspneic since start of third trimester, dyspnea much worse today. Syncopal episode at home.							
Past Medical History:		Single gestation pregnancy confirmed on 3 rd trimester U/S		Medications:		Prenatal vitamins	
Allergies: None							
Social History: No drugs, no EtOH, lives with husband and child. No psychiatric history.							
Family History: Non-contributory							
Review of Systems:		CNS:		Nil			
		HEENT:		Nil			
		CVS:		Peripheral edema, bilateral legs > arms.			
		RESP:		Dyspnea worsening over 3 rd trimester.			
		GI:		Nil			
		GU:		Pregnancy unremarkable. No vaginal bleeding or abdominal pain.			
		MSK:		Nil		INT:	
B. Baseline Simulator State and Physical Exam							
☐ No Monitor Display		☒ Monitor On, no data displayed			☐ Monitor on Standard Display		
HR: 110/min		BP: 90/50		RR: 35 /min		O ₂ SAT: 89% on RA	
Rhythm: Sinus rhythm		T:36.5 °C		Glucose: 5.8 mmol/L		GCS: 15 (E4 V5 M6)	
General Status: Moderate respiratory distress							
CNS:		Alert, oriented, PEARL 3mm, normal reflexes, no clonus					
HEENT:		Nil					
CVS:		Sinus rhythm with PVC's, peripheral edema 2+ to bilateral legs					
RESP:		Respiratory distress, 2-3 word sentences, crackles to bases					
ABDO:		Normal					
GU:		Fundus palpated at xyphoid, non-tender					
MSK:		Nil		SKIN:		Nil	

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tachycardia HR: 110/min BP: 90/50 RR: 35/min O ₂ SAT: 89% RA T: 36.5°C FHR: 150 bpm	Moderate respiratory distress, crackles on exam	<u>Learner Actions</u> - ☐ History & physical (ABC's) - ☐ IV x 2, O ₂ , monitors - ☐ Fetal heart monitor or check FHR with Doppler - ☐ Left lateral tilt with wedge - ☐ Glucose check - ☐ Labs – including trop, LFTs, and coags - ☐ CXR - ☐ ECG - ☐ POCUS (FAST & OB normal Cardiac shows poor EF, Lung shows B lines) - ☐ Call RT for BiPap - ☐ Call OB for urgent C/S - ☐ Trial 250-500ml IV NS bolus	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - No O ₂ placed: SpO ₂ to 85% - No lateral tilt: BP to 75/40 <u>Triggers</u> <i>For progression to next state</i> - All actions complete or 5 mins into case → 2. Respiratory failure
2. Respiratory failure Rhythm: Sinus tachycardia with PVC's HR: 115/min BP: 95/50 RR: 40/min, O ₂ SAT: 87% despite 100% FiO ₂ T: 36.5°C	Worsening respiratory distress, drowsy. +++crackles and wheeze bilaterally SpO ₂ starts dropping despite prior management	<u>Learner Actions</u> - ☐ Call RT - ☐ Call Anesthesia/ OB/ Cardiology/ NICU STAT - ☐ BVM with PEEP or NIPPV prior to intubation - ☐ Airway assessment (presume difficult airway) - ☐ RSI medications - ☐ Vasopressor at bedside (norepi, epinephrine, push dose phenylephrine) - ☐ Intubation + PEEP - ☐ Verbalize vent settings - ☐ Post-intubation sedation	<u>Modifiers</u> - If hypoxia is not noticed, RN to cue "I think she is becoming more drowsy and the sats are dropping. Should I call RT?" - PPV prior to intubation → SpO ₂ up to 95% <u>Triggers</u> - Intubation → 3. Hypotension

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Section VI: Scenario Progression

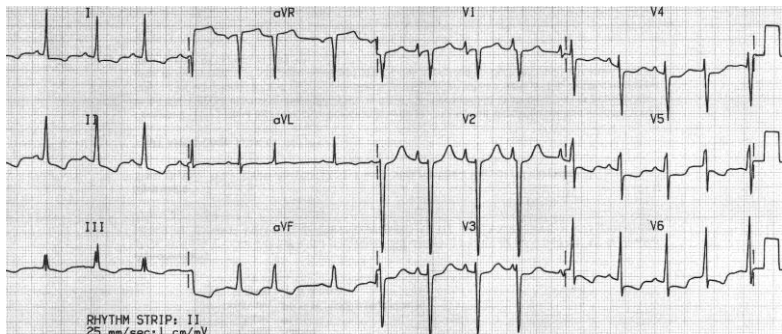
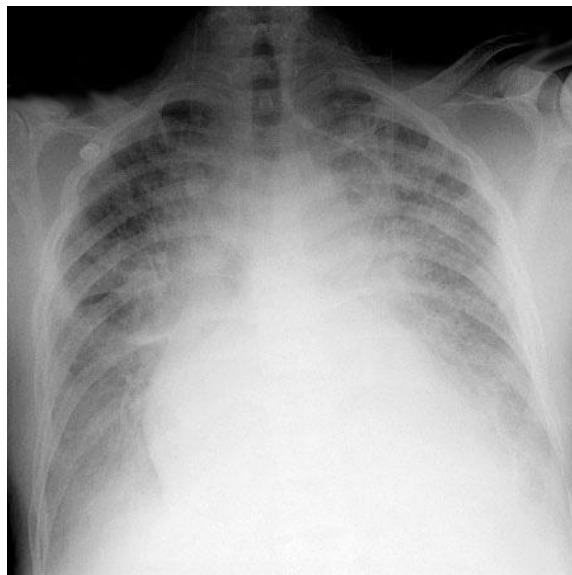
3. Hypotension Rhythm: sinus bradycardia HR: 100/min BP: 75/35 O2 sat: 95% with PEEP 10	Sedated Intubated Ventilated	<u>Learner Actions</u> - ☐ Consultants called (if not yet done) - ☐ Start vasopressor (Norepinephrine ideal) - ☐ Inotropes discussed +/- given (dobutamine 2.5-20 mcg/kg/min) - ☐ ± Set up for arterial line	<u>Modifiers</u> - Norepi started: BP to 80/40 - If no inotropes started, RN to cue: <i>"What do you think is going on here? Is this all her heart?"</i> <u>Triggers</u> - Inotropes started → 4. Improvement
4. Improvement Rhythm: sinus rhythm HR: 80/ min BP: 100/70 RR: 20 O2 sat: 95% on CMV	Sedated Intubated Ventilated	<u>Learner Actions</u> - ☐ Initiates inotropes and furosemide based on Cardiology's recommendations if not yet done - ☐ STAT formal Echo - ☐ Arterial line, central line (if not done) - ☐ Call OB again	<u>Modifiers</u> <u>Triggers</u> - Cardiology staff arrives after 3 mins → END CASE

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

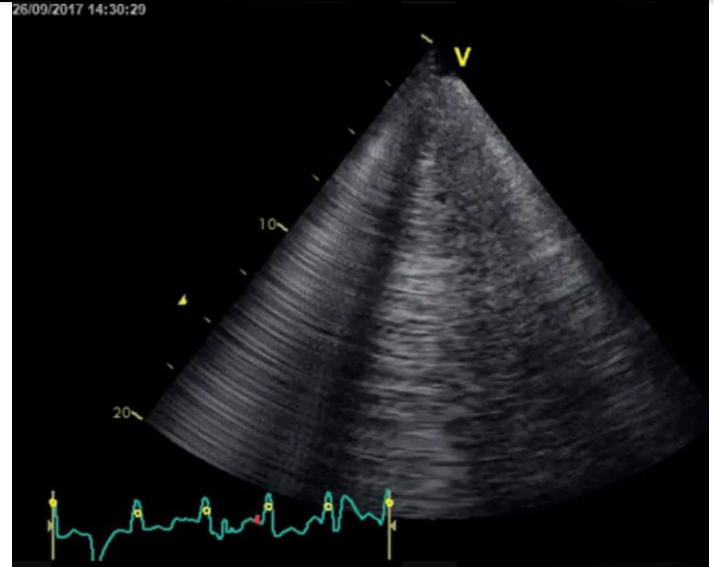
Laboratory Results						
Na: 140	K: 3.8	Cl: 109	HCO ₃ : 25	BUN:	Cr: 65	Glu: 5.8
Ca: 2.5		Mg: 1.0		PO ₄ : 0.90		Albumin: 35
VBG	pH: 7.20	PCO ₂ : 32	PO ₂ : 98	HCO ₃ : 23	Lactate: 2.8	
WBC: 16		Hg: 120		Hct: 0.370		Plt: 280
Trop: 60						

Images (ECGs, CXRs, etc.)	
 https://lifeinthefastlane.com/ecg-library/dilated-cardiomyopathy/	 https://www.med-ed.virginia.edu/courses/rad/cxr/postquestions/posttest.html
POCUS (apical 4 chamber view)- Impaired EF	POCUS Lung Views (Bilat)

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<http://www.thepocusatlas.com/echo/2hj4yjl0bcpxxokz-zzoyip9mnz1ck5>



<http://www.thepocusatlas.com/pulmonary/>

FAST- RUQ (still)



<http://sinaiem.us/tutorials/fast/us-ruq-normal/>

FAST- OB with FHR

https://www.youtube.com/watch?v=SKKnTLqI_VM

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Medical Objectives:	1. Recognize a critically ill pregnant patient with cardiogenic shock 2. Work through the differential diagnosis of dyspnea in pregnancy 3. Facilitate safe intubation with considerations for a pregnant patient
Sample Questions for Debriefing	
1. What are the physiological and anatomical considerations in the resuscitation of a pregnant patient? 2. Describe your differential for dyspnea in pregnancy 3. What is on the differential for etiologies of cardiogenic shock? How can you clinically differentiate between these on initial examination? 4. What are some risk factors for peripartum cardiomyopathy? 5. What considerations must be made in the airway management of a pregnant patient? What about RSI medications?	
Key Moments	
1. Recognizing and managing shock in a pregnant patient	
2. Intubation planning for the pregnant patient	
3. Providing inotropic support in a patient with cardiogenic shock	

